Kernix 0.12.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

51 workloads

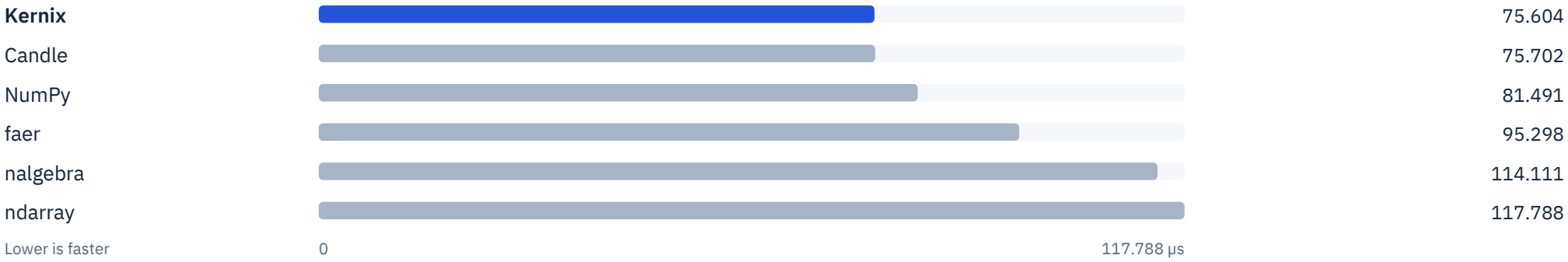
170 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.12.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



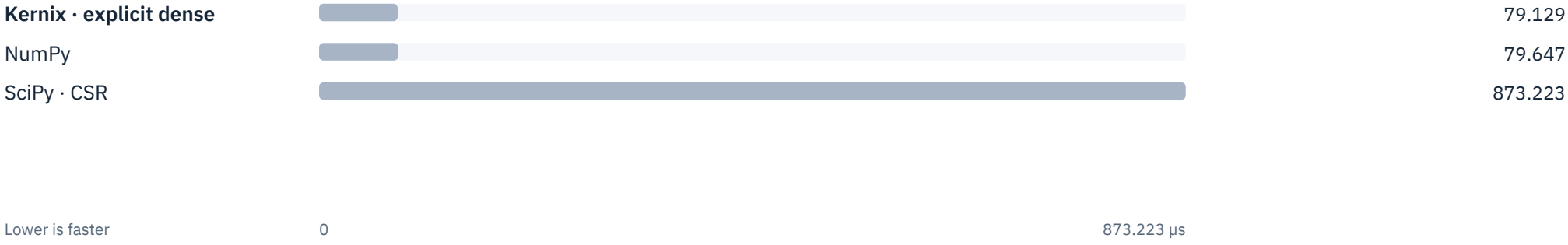
Kernix 0.12.0 / Sparse matrices & memory

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



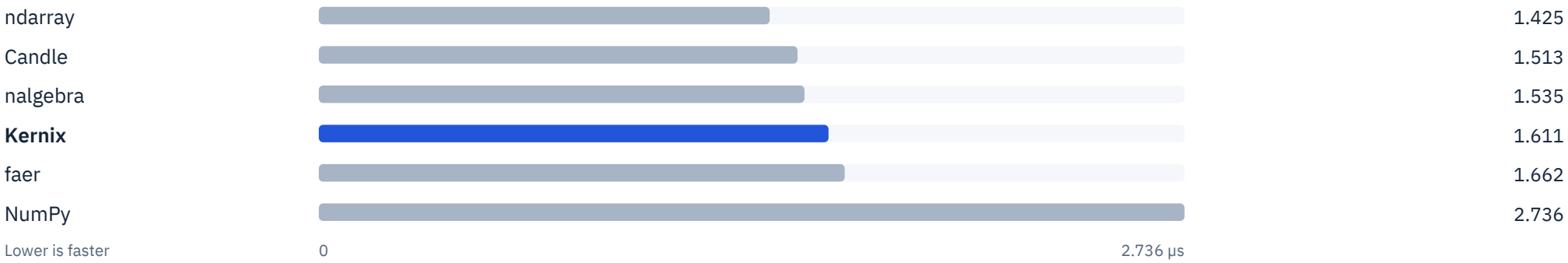
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

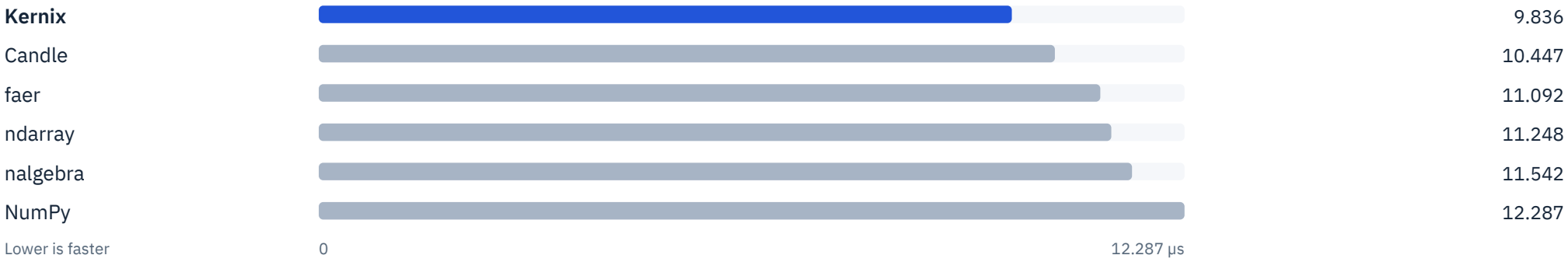
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



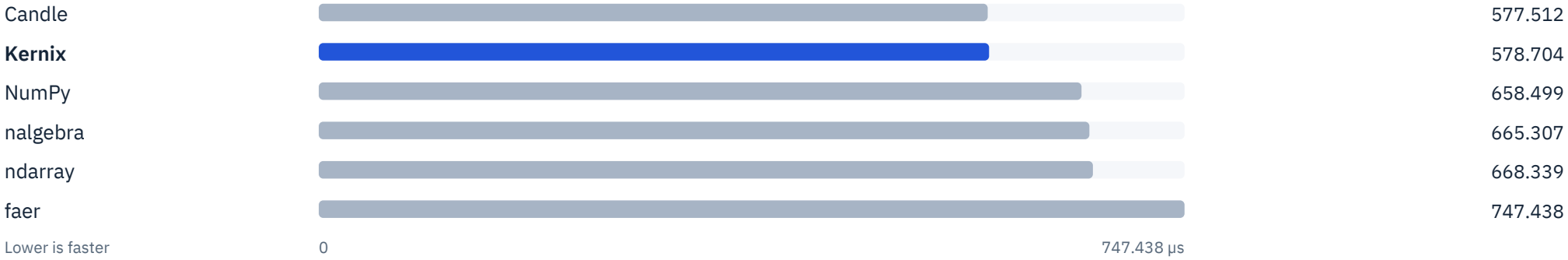
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



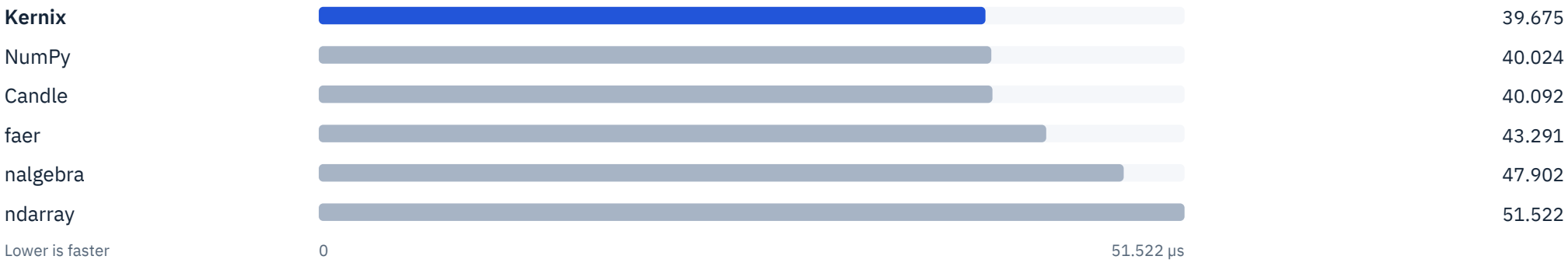
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

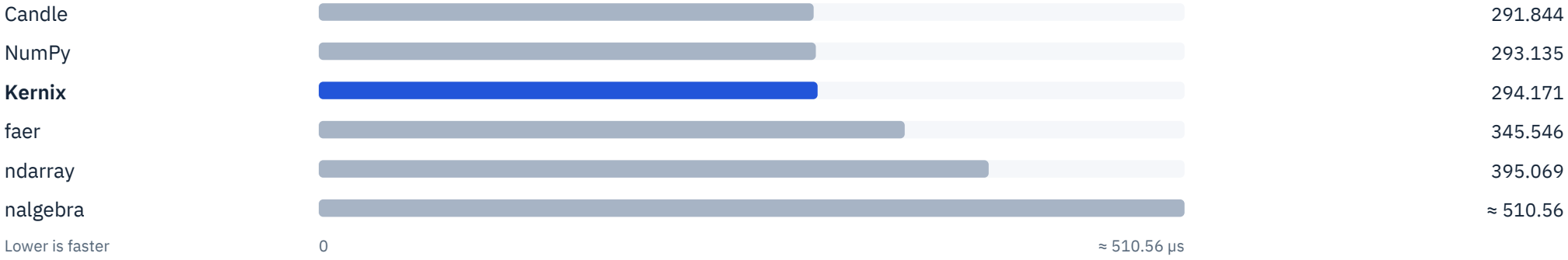
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



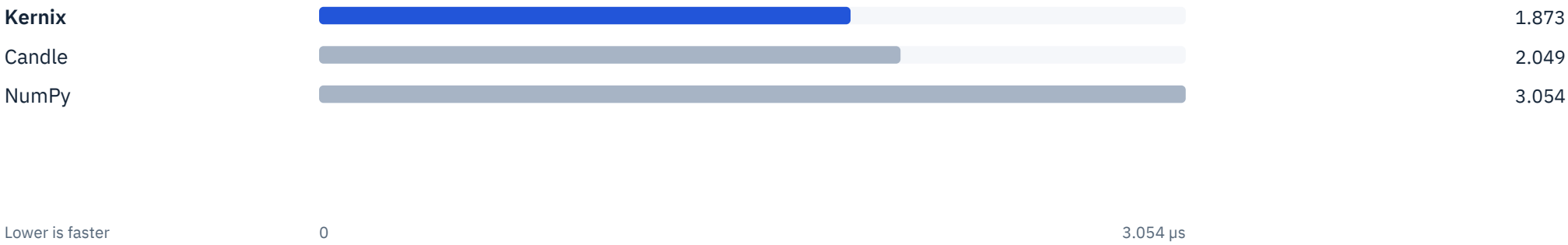
Kernix 0.12.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

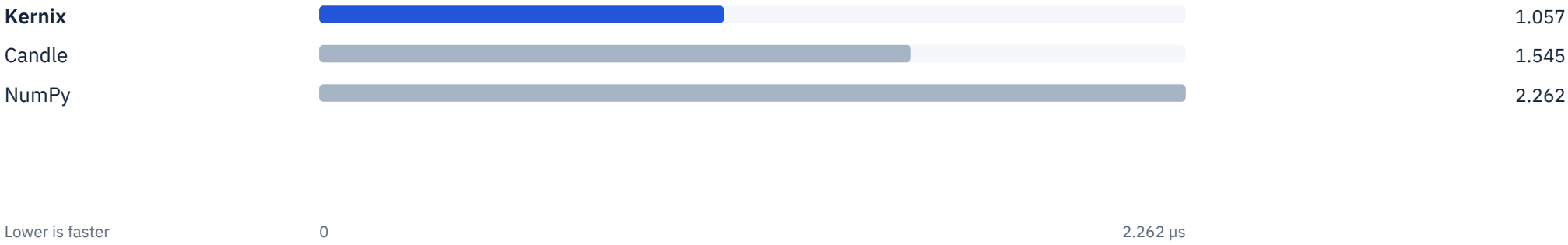
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

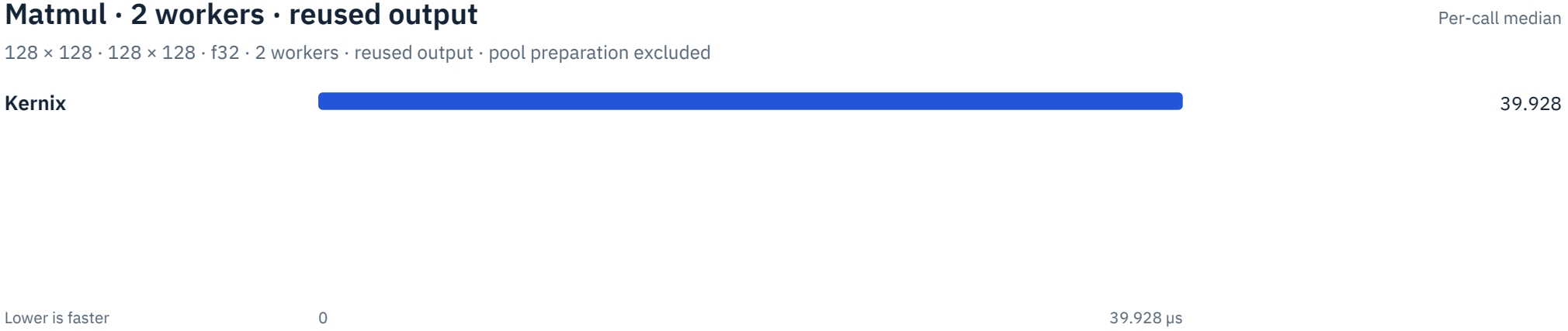
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



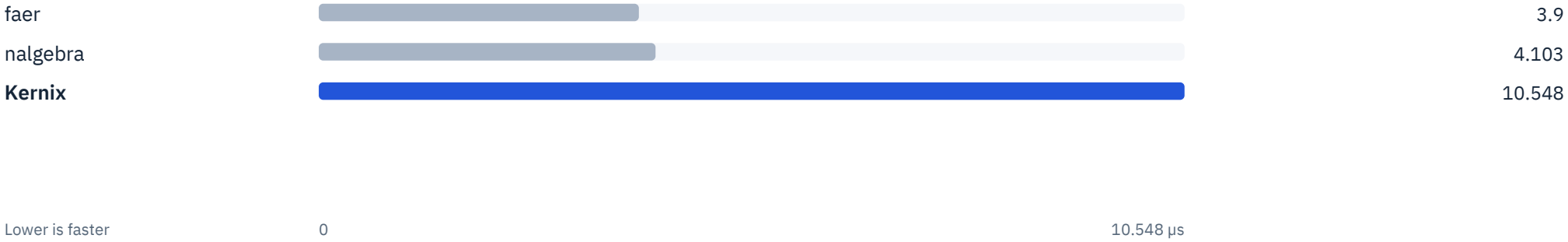
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



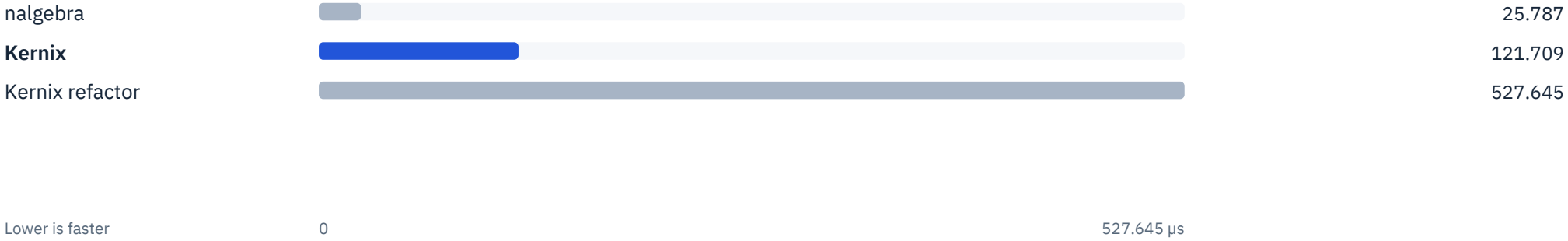
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

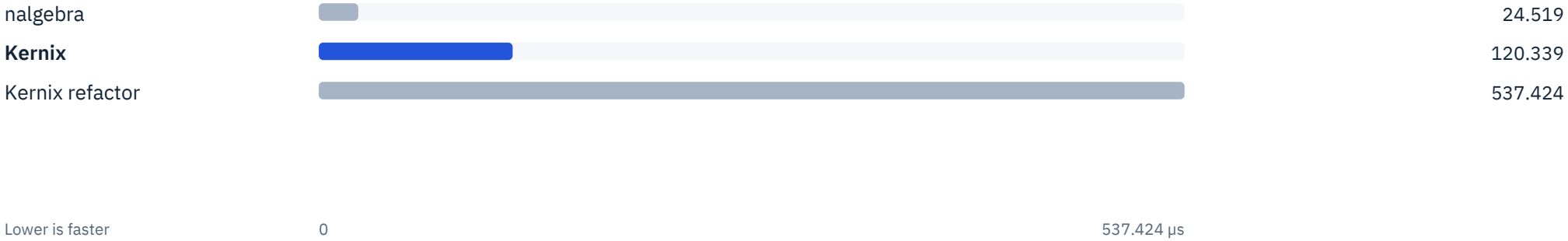
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



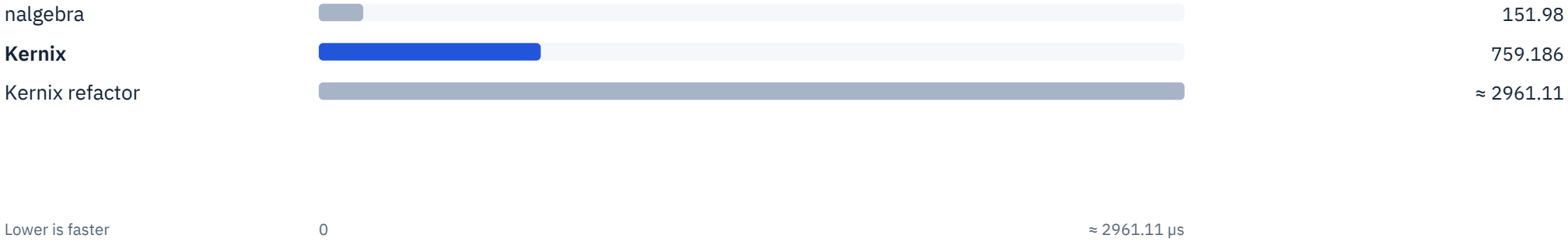
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

Per-call median

512 × 128 · f64 · 1 worker · allocated output



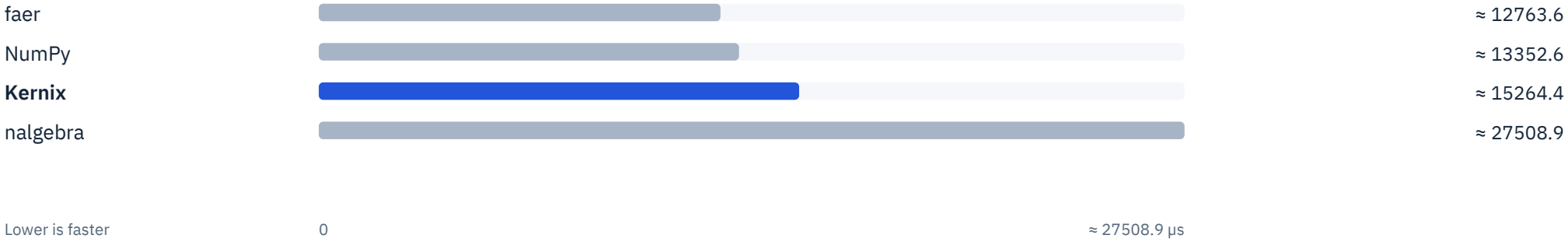
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

Per-call median

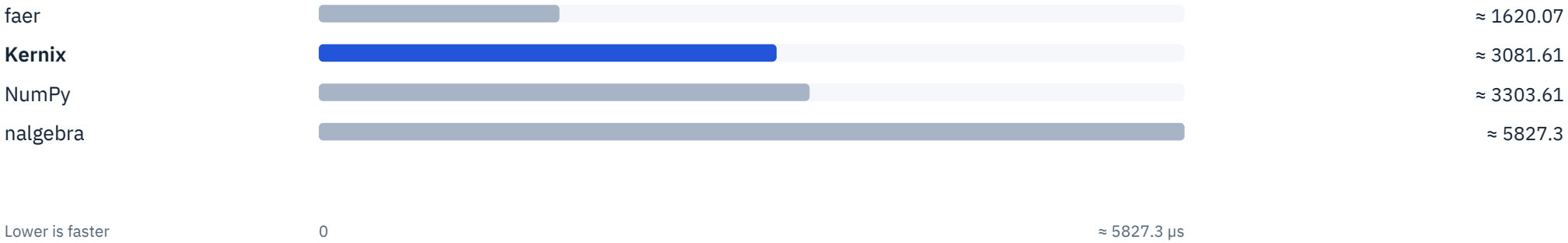
1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

512 × 512 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

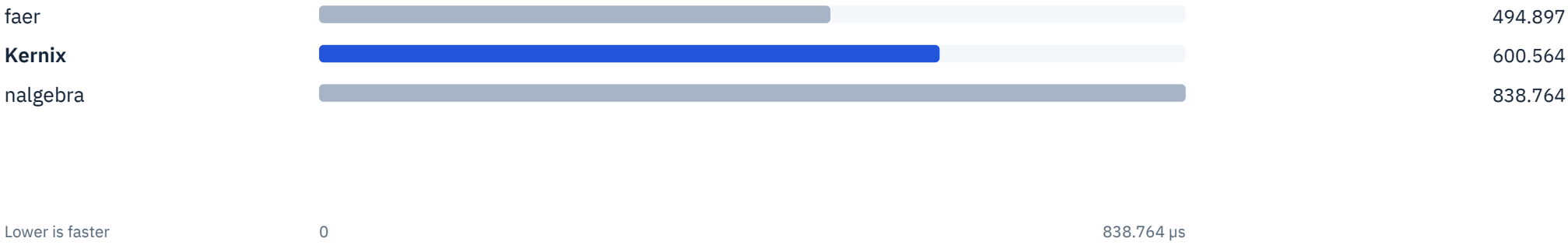
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



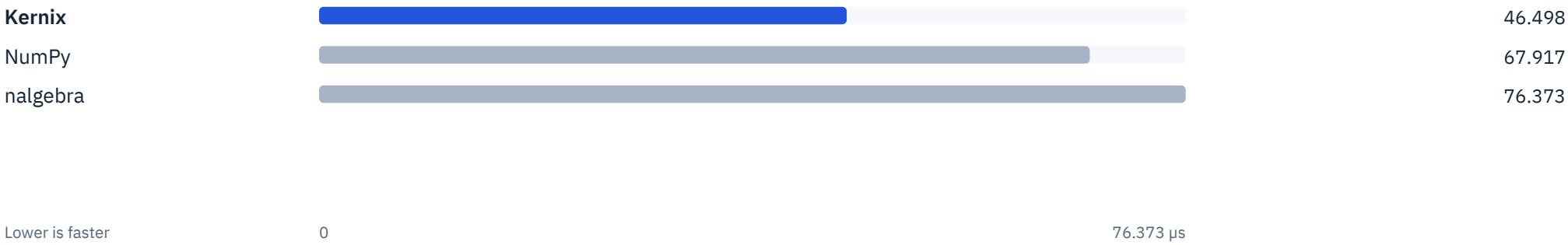
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

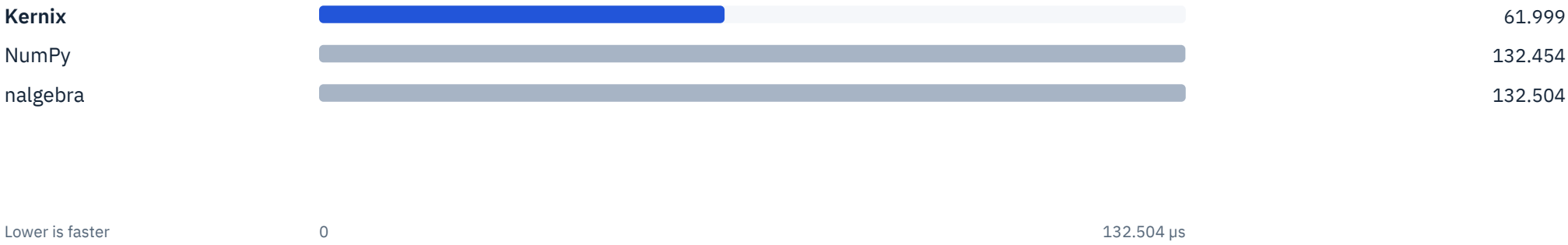
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



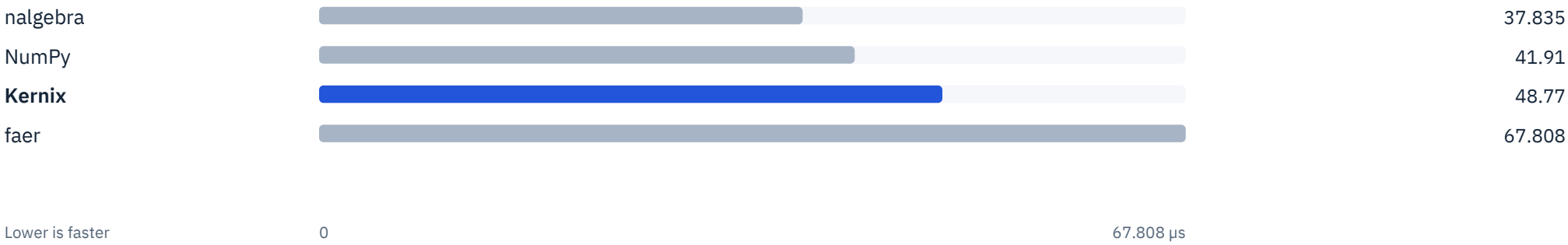
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



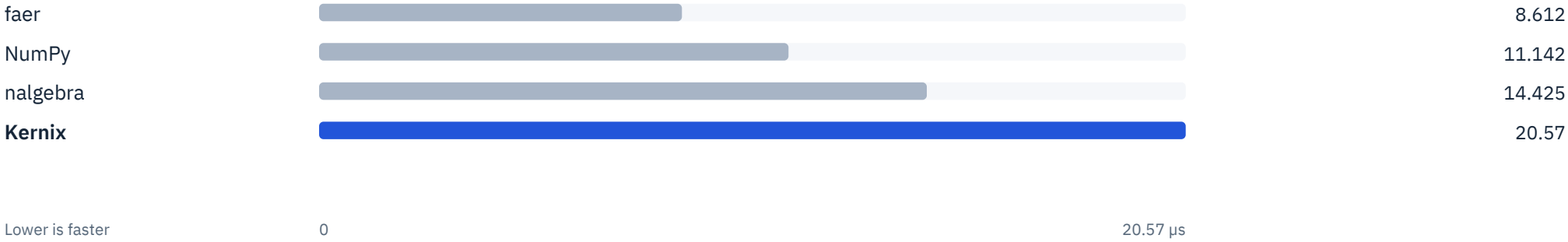
Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

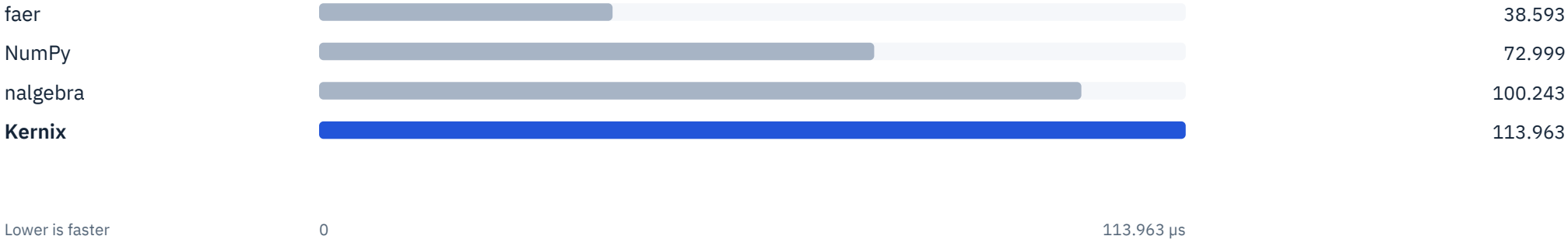
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



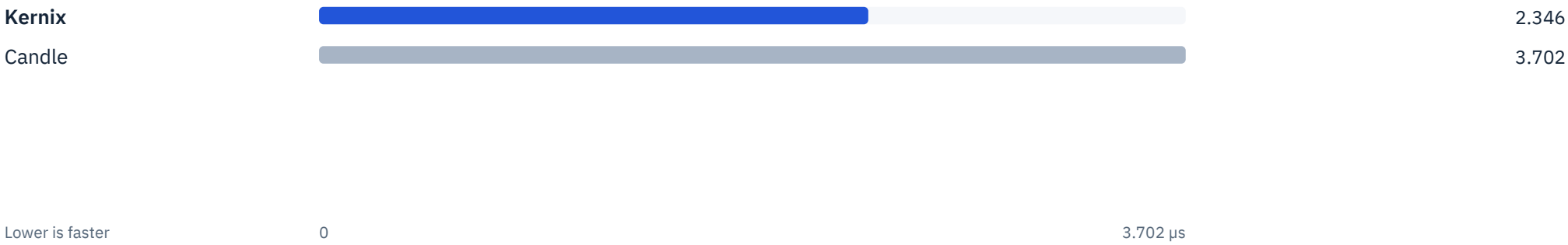
Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



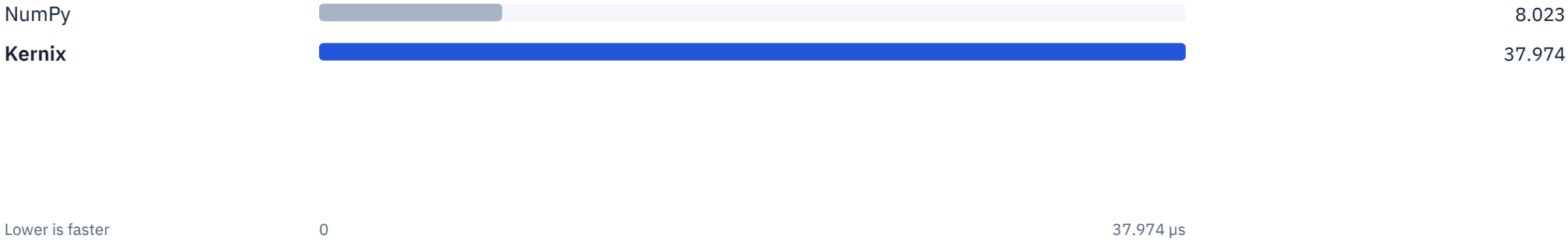
Kernix 0.12.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



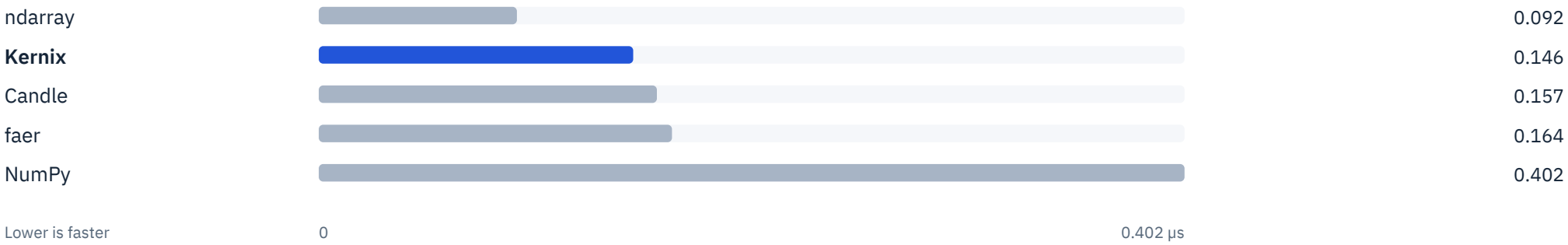
Kernix 0.12.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	7133	7.133
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	384024	384.024
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	4948	4.948
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	3230535	3230.535
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	20475	20.475
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	405344	405.344
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	381985	381.985
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	42102	42.102
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	14246	14.246
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	206305	206.305

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	39736	39.736
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	172879	172.879
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	43248	43.248
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	603759	603.759
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	574436	574.436
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	79647	79.647
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	873223	873.223
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	79129	79.129
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	84445	84.445
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	124804	124.804

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	49227	49.227
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	37672	37.672
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	78009	78.009
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	74251	74.251
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	35942	35.942
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	38065	38.065
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1662	1.662
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1535	1.535
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2736	2.736
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1425	1.425

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1513	1.513
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1611	1.611
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10447	10.447
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11542	11.542
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12287	12.287
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9836	9.836
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	11092	11.092
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11248	11.248
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	95298	95.298
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	117788	117.788

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	81491	81.491
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	114111	114.111
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	75604	75.604
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	75702	75.702
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	658499	658.499
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	577512	577.512
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	665307	665.307
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	578704	578.704
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	668339	668.339
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	747438	747.438

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5110	5.11
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4955	4.955
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6074	6.074
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6076	6.076
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	4971	4.971
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5698	5.698
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	39675	39.675
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	40092	40.092
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	40024	40.024
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	43291	43.291

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	47902	47.902
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51522	51.522
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	345546	345.546
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	294171	294.171
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	395069	395.069
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	510559.99999999994	510.55999999999994
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	293135	293.135
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	291844	291.844
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3054	3.054
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2049	2.049

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1873	1.873
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	1545	1.545
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1057	1.057
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2262	2.262
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	11917	11.917
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	40966	40.966
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	292801	292.801
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	39928	39.928
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	4103	4.103
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	10548	10.548

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	3900	3.9
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	227765	227.765
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	883708	883.708
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1093900	1093.9
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	7264539	7264.539
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	1561877	1561.877
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	8968499	8968.499
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	846889	846.889
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	241498	241.498
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1079062	1079.062

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	527645	527.645
Cholesky rank-one update / nalgebra 256 × 256 · f64	25787	25.787
Cholesky rank-one update / Kernix 256 × 256 · f64	121709	121.709
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	537424	537.424
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	24519	24.519
Cholesky rank-one downdate / Kernix 256 × 256 · f64	120339	120.339
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	2961108	2961.108
Cholesky rank-one update / Kernix 512 × 512 · f64	759186	759.186
Cholesky rank-one update / nalgebra 512 × 512 · f64	151980	151.98
Thin QR / Kernix 512 × 128 · f64	2682241	2682.241

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 512 × 128 · f64	2454159	2454.159
Thin QR / faer 512 × 128 · f64	2082605	2082.605
Thin QR / nalgebra 512 × 128 · f64	3066934	3066.934
Thin QR / nalgebra 1024 × 256 · f64	27508871	27508.871
Thin QR / Kernix 1024 × 256 · f64	15264449	15264.449
Thin QR / faer 1024 × 256 · f64	12763563	12763.563
Thin QR / NumPy 1024 × 256 · f64	13352622	13352.622
Cholesky factorization / faer 512 × 512 · f64	1620073	1620.073
Cholesky factorization / Kernix 512 × 512 · f64	3081611	3081.611
Cholesky factorization / nalgebra 512 × 512 · f64	5827295	5827.295

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 512 × 512 · f64	3303614	3303.614
Column-pivoted QR / faer 96 × 32 · f64	77223	77.223
Column-pivoted QR / Kernix 96 × 32 · f64	57371	57.371
Column-pivoted QR / nalgebra 96 × 32 · f64	75452	75.452
Column-pivoted QR / Kernix 256 × 64 · f64	600564	600.564
Column-pivoted QR / faer 256 × 64 · f64	494897	494.897
Column-pivoted QR / nalgebra 256 × 64 · f64	838764	838.764
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	46498	46.498
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	67917	67.917
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	76373	76.373

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	132454	132.454
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	61999	61.999
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	132504	132.504
Thin QR / Kernix 96 × 32 · f64	48770	48.77
Thin QR / faer 96 × 32 · f64	67808	67.808
Thin QR / NumPy 96 × 32 · f64	41910	41.91
Thin QR / nalgebra 96 × 32 · f64	37835	37.835
Thin QR / nalgebra 256 × 64 · f64	384992	384.992
Thin QR / NumPy 256 × 64 · f64	357744	357.744
Thin QR / faer 256 × 64 · f64	421185	421.185

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 256 × 64 · f64	495985	495.985
Cholesky factorization / NumPy 64 × 64 · f64	11142	11.142
Cholesky factorization / Kernix 64 × 64 · f64	20570	20.57
Cholesky factorization / nalgebra 64 × 64 · f64	14425	14.425
Cholesky factorization / faer 64 × 64 · f64	8612	8.612
Cholesky factorization / nalgebra 128 × 128 · f64	100243	100.243
Cholesky factorization / NumPy 128 × 128 · f64	72999	72.999
Cholesky factorization / faer 128 × 128 · f64	38593	38.593
Cholesky factorization / Kernix 128 × 128 · f64	113963	113.963
Cholesky factorization / faer 256 × 256 · f64	208438	208.438

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	690745	690.745
Cholesky factorization / NumPy 256 × 256 · f64	530914	530.914
Cholesky factorization / Kernix 256 × 256 · f64	506734	506.734
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6630	6.63
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	3934	3.934
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2369	2.369
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4397	4.397
Model inference / Candle batched normalized/sliced MLP · f64	3702	3.702
Model inference / Kernix batched normalized/sliced MLP · f64	2346	2.346
Model recording + forward / Candle batched normalized/sliced MLP · f64	5252	5.252

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2665	2.665
Model backward / Kernix prepared graph → fresh gradients · f64	3337	3.337
Model backward / Candle prepared graph → fresh gradients · f64	14785	14.785
Parameter update / Kernix transactional SGD, fixed gradients · f64	527	0.527
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	37974	37.974
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	8023	8.023
Row softmax / Candle 256 × 64 · f64	84045	84.045
Row softmax / Kernix 256 × 64 · f64	89846	89.846
Row softmax / NumPy 256 × 64 · f64	101847	101.847
Row softmax / ndarray 256 × 64 · f64	74404	74.404

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	402	0.402
Create zeros / ndarray 64 × 64 · f32	92	0.092
Create zeros / faer 64 × 64 · f32	164	0.164
Create zeros / Candle 64 × 64 · f32	157	0.157
Create zeros / Kernix 64 × 64 · f32	146	0.146
Create ones / Candle 64 × 64 · f32	255	0.255
Create ones / faer 64 × 64 · f32	199	0.199
Create ones / ndarray 64 × 64 · f32	147	0.147
Create ones / Kernix 64 × 64 · f32	184	0.184
Create ones / NumPy 64 × 64 · f32	1342	1.342